AquaSentinel: Explainable Water Potability Prediction

R-Based Classification, Model Evaluation & Local Explainability

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Conclusion & Summary

This project implemented an end-to-end water-potability classification workflow in R using the Water Potability dataset from Kaggle. The dataset contains **3,276 records**, nine water-quality predictors and the target variable *Potability*. Initial inspection identified missing values in Sulfate (781), pH (491) and Trihalomethanes (162), with no exact duplicate rows. Missing predictor values were imputed using medians calculated from the training split only. The data was divided into training (2,292), validation (490) and test (494) records.

Logistic Regression was used as the baseline classifier and Random Forest as a comparison model. Random Forest was selected because its validation ROC-AUC (**0.6272**) exceeded Logistic Regression (0.4366). On the held-out test set, Random Forest achieved **67.21% accuracy**, **65.05% precision**, **34.72% recall**, **45.27% F1-score** and **0.6910 ROC-AUC**. The confusion matrix recorded 265 true negatives, 36 false positives, 126 false negatives and 67 true positives.

The results show some class-discrimination ability, but the low recall means that many test samples labelled potable were missed at the 0.5 threshold. LIME was used to examine Case 968: the predicted label was 0, with P(Potable = 1) = 0.1833 and a local surrogate fit of **R^2 = 0.134**. Because this local fit is low, its feature contributions should be interpreted cautiously. Future work should examine threshold calibration using validation data, strategies to improve potable-class recall, and evaluation on additional representative data. AquaSentinel predicts dataset labels; it does not certify drinking-water safety or replace laboratory and regulatory testing.

Model Comparison (Validation Split)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	61.22%	100.00%	0.52%	1.04%	0.4366
Random Forest (selected)	**62.24%**	**53.19%**	**26.18%**	**35.09%**	**0.6272**

Selected Model Performance (Held-Out Test Split)

Accuracy	Precision	Recall	F1-score	ROC-AUC
67.21%	**65.05%**	**34.72%**	**45.27%**	**0.6910**

Project Links & Metadata

1. Project Repository	https://github.com/Aziz1034/AquaSentinel
2. Dataset Source	Kaggle - Water Potability Dataset
3. Dataset URL	https://www.kaggle.com/datasets/adityakadiwal/water-potability
4. R Notebook	AquaSentinel Review 2 - Google Colab

Interpretation note: Low potable-class recall (34.72%) and the LIME local fit (R^2 = 0.134) are important limitations. The model requires further validation and must not be used as a standalone drinking-water safety decision tool.

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Academic Year 2026-27